MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8165

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode
See figure 1
Mounting position: Vertical, base down or up
Weight: 3-ounces nominal (85 grams)

ABSOLUTE RATINGS: C Telegraphy

Parameter:	Fl	Ef	Eb	Ec1	Ec2	Ib	Pp	Pg1	Pg2	T(seal)	TE
Unit:	MHz	Vac	Vdc	Vdc	Vdc	mAdc	W	W	W	°C	°C
Maximum:	150	6.3	3,000	-500	400	150	65	5	10	200	225
Minimum:	---	5.7	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	---	6.0	1,000	Adj	400	65	---	---	---	Note 2	---

GENERAL:
Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	3.2	3.8	Aac
1231	Peak emission	-	eb = ec1 = ec2 = 2,500 v	0.65	II	is	2.0	---	a
1256	Electrode current (screen)	-		0.65	II	Ic2	-1.0	+1.0	mAdc
1261	Electrode voltage (grid)	-		0.65	II	Ec1	-38.0	-58.0	Vdc
1266	Total grid current	1		0.65	II	Ic1	---	-5	µAdc
1266	Primary grid emission (control)	-	Ef = 6.6 Vac; Pg1 = 6 W or Ic1 = 28 mAdc; t = 15; anode and screen-grid floating	0.65	II	Isg1	---	-250	µAdc
1266	Primary grid emission (screen)	-	Ef = 6.6 Vac; Pg2 = 12 W or Ic2 = 27 mAdc; Ec1 = 0; t = 15; anode floating	0.65	II	Isg2	---	-250	µAdc
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltages	---	---	---	---	---	---
1036	Bump	-	Angle = 20°	---	---	---	---	---	---
1036	Bump and short	-	Angle = 15°	---	---	---	---	---	--
1316	Amplification factor	-	g1 to g2; Ec2 = 250 Vdc; Ic2 = 40 mAdc; Eb = 0	---	---	---	---	---	---
1331	Direct-interelectrode capacitance	-		---	---	Cgp	---	0.12	pF
						Cin	6.0	8.3	pF
						Cout	1.9	2.6	pF
(E) 1236	Power oscillation	-	Eb = 1,500 Vdc; Ib = 150 mAdc; Ec2 = 250 Vdc; F = 150 MHz	---	---	Po	110	---	W
	Quality conformance inspection, part 3								
---	Life test	-	Group C; power oscillation; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
1231	Peak emission	-		---	---	is	1.6	---	a
1266	Primary grid emission (control)	-		---	---	Isg1	---	-250	µAdc
1266	Primary grid emission (screen)	-		---	---	Isg2	---	-250	µAdc

NOTES:
1. This test is to be the first test performed at the conclusion of the holding period.
2. During all electrical tests involving application of filament power, a heat-dissipating connector (Eimac HR-6, or equivalent) may be used on the anode terminal.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 17, 80
DSA - ES

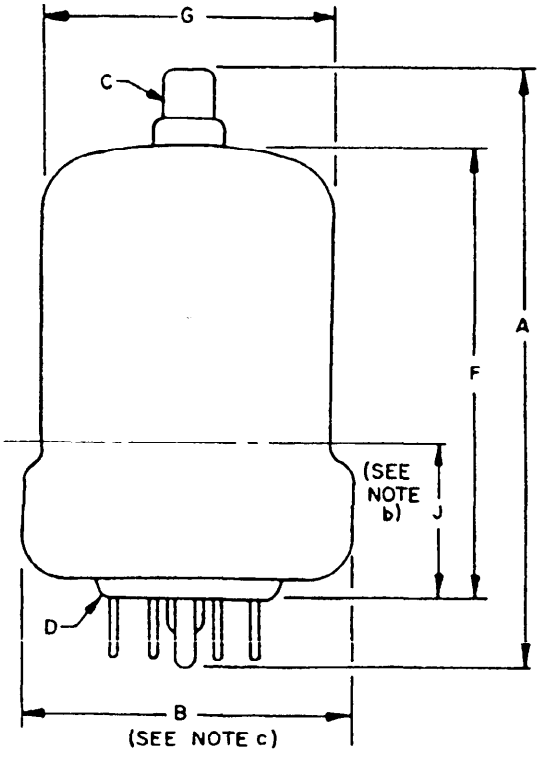
User activities:
Army - ME, WC
Navy - AS, OS, MC, CG, SH
Air Force - 11, 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2828)

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Ⓔ

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Quality conformance inspection, part 2	
A	4.000 (101.60)	4.375 (111.13)
B		2.375 (60.33)
G		2.060 (52.32)
J	.844 (21.44)	1.219 (30.96)
	Quality conformance inspection, part 3 (periodic check)	
C	Cap: C1-22 (EIA)	
D	Base: E5-59 (see note b) (EIA)	
F	2.938 (74.63)	3.313 (84.15)

PIN CONNECTIONS	
Pin No.	Element
1	f
2	g2
4	g1
6	g2
7	f
cap	a

NOTES:

Ⓔ a. For pin alinement use JEDEC gage GE7-6. A molded glass flare base with pin configuration identical with that of base E5-59 may be used.

Ⓔ b. With base pins inserted into gage GE7-6, a flat plate gage with a 2.063 ± .003 (52.40 ± .08) diameter hole is passed over diameter G until it stops at position defined by dimension J. Dimension J shall then be gaged between bottom surface of flat-plate gage and top surface of gage GE7-6 with surfaces of both gages parallel. The 2.063 (52.40) diameter hole of the flat-plate gage and .500 (12.70) diameter hole of gage GE7-6 shall be concentric within .150 (3.81).

c. A ring gage .125 ± .010 (3.18 ± .26) thick and 2.125 (53.98) minimum and 2.128 (54.05) maximum inside diameter shall not pass over diameter B when tried at any angle.

FIGURE 1. Outline drawing of electron tube type 8165.

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